Ruiqi Rae Yu

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Google Scholar

Professional Experience

Quantitative Researcher 2026 – Present
Chicago Trading Company

Education

Ph.D. in Operations Research & Financial Engineering 2021 – 2026
Princeton University

- Topics: Mathematical Statistics, Causal Inference
- Advisor: Matias Cattaneo, Department of Operations Research & Financial Engineering

M.A. in Operations Research & Financial Engineering 2021 – 2023
Princeton University

B.Sc. in Mathematics with Honors 2017 – 2021
University of Toronto

Research & Publications

Publications

1. Estimation and Inference in Boundary Discontinuity Designs: Distance-Based Methods, with M. D. Cattaneo and R. Titiunik.
   Journal of Econometrics, *256* (Part A), Article 106266, 2026.
2. Strong approximations for empirical processes indexed by Lipschitz functions, with M. D. Cattaneo.
   The Annals of Statistics, *53*(3), 1203–1229, 2025.

Working Papers

1. Estimation and Inference in Boundary Discontinuity Designs: Location-Based Methods, with M. D. Cattaneo and R. Titiunik.
   Journal of the American Statistical Association, minor revision.
2. Accuracy Limits of Causal Trees for Individualized Treatment Effects, with M. D. Cattaneo and J. M. Klusowski.
   Proceedings of the National Academy of Sciences, revision requested.
3. Robust Inference for the Direct Average Treatment Effect with Treatment Assignment Interference, with M. D. Cattaneo and Y. He.
   arXiv preprint, 2025.
4. Estimation and Inference in Boundary Discontinuity Designs: Pooling-Based Methods, with M. D. Cattaneo and R. Titiunik. *Work in progress.*

Book Chapters

1. Boundary Discontinuity Designs: Theory and Practice, with M. D. Cattaneo and R. Titiunik.
   Advances in Economics and Econometrics: Thirteenth World Congress, Cambridge University Press, Vol. 1, Ch. 2, forthcoming.

Software Articles

1. rd2d: Causal Inference in Boundary Discontinuity Designs, with M. D. Cattaneo and R. Titiunik.
   arXiv preprint, revised 2026.
   Companion software for R, Python, and Stata: rd2d.

Teaching

Princeton University

- ORF 524: Statistical Theory and Methods — Fall 2025
- ORF 245: Fundamentals of Statistics — Fall 2022, Spring 2023, Fall 2024

University of Toronto

- MAT 244: Ordinary Differential Equations — Fall 2020
- MAT 137: Calculus — Spring 2019
- MAT 223: Linear Algebra — Fall 2018

Awards

Princeton University

- DDSS Graduate Fellowship in Social Data Science — 2024, 2025

University of Toronto

- Coxeter Scholarship in Mathematics — 2021, 2019
- Dean's List Scholar — 2020, 2019, 2018
- The Chancellor's Scholarships — 2020
- The George Gray Falle Scholarships — 2019, 2018
- University of Toronto Scholar – Rogers — 2019
- Adel S Sedra Undergraduate Scholarship in Mathematics — 2018

Services

Peer Review

- Economic Letters
- Econometric Theory
- Journal of Causal Inference
- Journal of Econometrics
- Journal of the American Statistical Association
- Journal of the Royal Statistical Society Series A
- Operations Research
- Statistical Science

Presentations & Conferences

- Joint Statistical Meeting — 2025
  Invited Paper Session on Regression Discontinuity Design with Complex Data
- Data-Driven Social Science Connect at Princeton University — 2024